

MEHWISH MALIK

AI Engineer | Agentic AI Developer | Full-Stack Developer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer with hands-on experience building agentic AI systems, Retrieval-Augmented Generation (RAG) applications, and production-ready web platforms. Skilled across the modern AI stack — Python, FastAPI, TypeScript, and Next.js — with practical experience in LLM API integration, vector search, database design, and containerization fundamentals (Docker, Kubernetes). Proven ability to independently ship full-stack products end-to-end, from backend architecture to polished user interfaces. Passionate about building intelligent agents and AI-driven products that solve real problems.

CORE SKILLS

AI & Agentic Engineering: Agentic AI, RAG (Retrieval-Augmented Generation), LLM API Integration, Prompt Engineering, Multi-Agent Systems, MCP, AI-Driven Development, Spec-Driven Development

Backend & Data: Python, FastAPI, REST API Design, SQLite, Database Design

Frontend & Full-Stack: TypeScript, JavaScript, Next.js, React, Tailwind CSS, HTML5, CSS3

DevOps & Tools: Docker (fundamentals), Kubernetes (fundamentals), Git, GitHub, Claude Code, Streamlit, Sanity CMS

Professional: Problem Solving, Agile Development, Team Collaboration, Technical Communication

KEY PROJECTS

RAG Chatbot | *FastAPI, Python, SQLite, LLM API*

- Built a Retrieval-Augmented Generation chatbot with a FastAPI backend and SQLite-based vector search enabling context-aware, grounded question answering without an external vector database.
- Designed the retrieval + generation pipeline end-to-end: document ingestion, embedding storage, similarity search, and LLM-based response generation.

AI Humanoid Robotics Book | *Claude Code, Spec-Driven Development, Next.js*

- Authored a comprehensive digital book on humanoid robotics using spec-driven development and AI-assisted prompting workflows, covering robotics fundamentals, AI integration, and real-world applications.

Secure Data Encryption System | *Python, Streamlit, Cryptography*

- Built a Python encryption tool with a Streamlit interface implementing industry-standard cryptographic algorithms for secure data protection and key management.

Task Management Application | *Next.js, TypeScript, Authentication, Database*

- Developed a full-stack task manager with user authentication and persistent, organized task workflows.

E-Commerce Platform & Blog Platform | *Next.js, Sanity CMS, Tailwind CSS*

- Built a responsive e-commerce storefront and a headless-CMS blog platform, both featuring dynamic content management and SEO-optimized, production-ready UI.

EXPERIENCE

Student Developer — GIAIC (Governor's Initiative for AI & Computing)

2024 – Present

- Built multiple production-oriented projects spanning AI engineering, agentic workflows, and full-stack web development.
- Integrated third-party APIs, CMS platforms, and databases into responsive, production-ready applications.
- Applied Python and Agentic AI concepts to design autonomous, task-oriented systems.
- Participated in hackathons and collaborative development sprints within an agile team environment.

EDUCATION

- Bachelor of Commerce (B.Com) — University of Karachi
- Intermediate (I.Com) — Degree College for Women
- Matriculation — Modern Public Secondary School

LANGUAGES

English, Urdu